✅ **STEP 1: Import the Dataset into Power BI**

1. Open **Power BI Desktop**
2. Click **Home → Get Data → Text/CSV**
3. Select your file: **ProjectManagementData.csv**
4. Click **Load**

Your dataset will appear as a table in Power BI.

✅ **STEP 2: Data Cleaning (Power Query Editor)**

Go to **Home → Transform Data**.

Check the following:

- **Date columns** must be in date format (StartDate, EndDate)
- **Hours and Cost columns** numeric
- Replace nulls in ActualHours and ActualCost with **0**
    - Right-click column → Replace Values → null → 0

Optional: Add useful calculated columns in Power Query (no DAX)

1️⃣ **Task Duration (Days)**

Duration = Duration.Days([EndDate] - [StartDate])

2️⃣ **Cost Variance**

CostVariance = [ActualCost] - [EstimatedCost]

3️⃣ **Hours Variance**

HoursVariance = [ActualHours] - [EstimatedHours]

4️⃣ **Task Health**

TaskHealth = if [ActualCost] > [EstimatedCost] then "Over Budget" else "Within Budget"

Close & Apply.

✅ **STEP 3: Create Relationships (Optional)**

If you only have 1 table, no relationship is required.

If you add future tables (managers, projects), connect them using:

- ProjectID

- Manager
- Department

✅ STEP 4: Create Key Visuals for Project Management Dashboard

Below are the best visualizations for project management.

📌 1. Project Summary KPI Cards

Go to **Visualizations → Card**

Add these fields:

- Total Projects → COUNT DISTINCT ProjectID
- Total Tasks → COUNTROWS
- Completed Tasks → filter TaskStatus = Completed
- In-Progress Tasks
- Total Estimated Cost → SUM(EstimatedCost)
- Total Actual Cost → SUM(ActualCost)

📌 2. Task Status Distribution (Pie or Donut Chart)

Visual: **Donut Chart**
Fields:

- Legend: TaskStatus
- Values: Count of TaskName

This shows how many tasks are:

- Completed
- In Progress
- Not Started

📌 3. Project Cost Analysis (Bar Chart)

Visual: **Clustered Bar Chart**
Fields:

- Axis: ProjectName
- Values: SUM(EstimatedCost) and SUM(ActualCost)

You will clearly see:

- Which projects are over budget
- Actual vs estimated comparison

📌 4. Hours Variance Chart

Visual: **Clustered Column Chart**
Fields:

- Axis: ProjectName
- Values: SUM(EstimatedHours), SUM(ActualHours)

Shows resource overuse or underuse.

📌 5. Risk Level by Project

Visual: **Stacked Column Chart**
Fields:

- Axis: ProjectName
- Legend: RiskLevel
- Values: Count of TaskName

This helps identify high-risk projects.

📌 6. Project Timeline (Gantt Chart)

To create a Gantt chart:

1. Go to **Visualizations → Get more visuals**
2. Search for **Gantt Chart (Microsoft)** → Add
3. Use:
    - Task: TaskName
    - Project: ProjectName
    - Start Date: StartDate
    - End Date: EndDate
    - Duration: Duration
    - % Completion:
        - Completed = 100

        - In Progress = 50
        - Not Started = 0

This gives you a project timeline view.

📌 7. Manager Performance (Bar or Matrix)

Visual: **Matrix**

- Rows: Manager
- Columns: TaskStatus
- Values: Count of TaskName

You immediately see:

- Who manages the most tasks
- Who completes the most
- Who has delays

📌 8. Department-Wise Cost Analysis

Visual: **Treemap**

- Group = Department
- Values = SUM(ActualCost)

Shows which department spends the most.

📌 9. Project Over-Budget Indicator

Visual: **Table + Conditional Formatting**

Fields:

- ProjectName
- EstimatedCost
- ActualCost
- CostVariance
- TaskHealth

Add conditional formatting:

- Highlight CostVariance in red if > 0
- Highlight TaskHealth:

- Red = Over Budget
- Green = Within Budget

📌 10. Project Completion Report

Visual: **Bar Chart or KPI**

- Axis: ProjectName
- Value: COUNT(TaskName) where TaskStatus = Completed

Helps quickly see which project is progressing faster.

✅ STEP 5: Build an Executive Dashboard Page

Your dashboard layout might look like this:

TOP ROW

- KPI Cards: Total Projects, Total Tasks, Completed Tasks, Total Cost

MIDDLE ROW

- Cost Comparison (Bar Chart)
- Hours Variance (Column Chart)

BOTTOM ROW

- Gantt Chart
- Project Risk Chart
- Manager Performance Table

This creates a powerful management-level dashboard.

✅ STEP 6: Insights You Can Present to Executives

Using your dataset, Power BI will show insights like:

✓ Which projects are over budget

✓ Where tasks are delayed

✓ High-risk projects

✓ Manager productivity

✓ Department with highest spending

✓ Real-time progress vs estimation

✓ **Resource-overused projects**

✓ **Timeline issues (Gantt)**

Executives can quickly see:

- Cost variances
- Performance bottlenecks
- Risk areas
- Workload distribution
- Project delays